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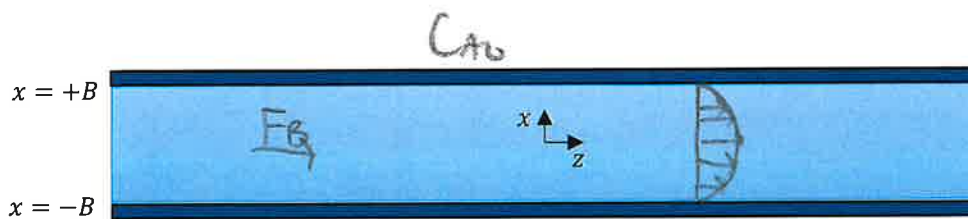
Joshua Olusiji

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TRANSPORT PHENOMENA II, CHE 312, SPRING 2025

FINAL EXAM

CLOSED BOOK, TIME: 120 MINUTES



The above schematic and the reference frame apply to both problems.

1. Liquid B is laminarly flowing between two flat walls placed a distance $2B$ apart, with the velocity profile

$$v_z = v_{z,max} \left[1 - \left(\frac{x}{B} \right)^2 \right]$$

The solid walls are made of a species A that is slightly soluble in the liquid B . Species A diffuses into the liquid according to Fick's law for binary diffusion with a constant diffusivity D_{AB} , where it undergoes a chemical reaction according to a first order rate law $R_A = -k_1''' c_A$.

Answer the questions in terms of the following parameters (all assumed constant):

$v_{z,max}$: maximum velocity of the fluid in z -direction

B : half the distance between the two parallel walls

D_{AB} : Fick's law diffusivity of A in B

k_1''' : rate constant in units of inverse time.

$$R_A = -k_1''' c_A$$

- 1.1. By setting up a shell mole balance, derive a partial differential equation for the concentration of species A as a function of x and z . Assume the concentration of A is very small $x_A \ll 1$, and neglect the diffusive transport of A in z direction.

$$\int_V dV \frac{\partial}{\partial t} c_A = - \int_S dS \langle N_A | n \rangle + \int_V dV R_A$$

- 1.2. Sufficiently far downstream, the concentration profile c_A is independent of z . Find this asymptotic concentration profile as a function of x , assuming the concentration of A at both walls is c_{A0} .

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2. Very thick slurries and pastes sometimes move in a channel almost as a solid plug with a uniform velocity profile. Consider a fluid between two flat walls placed a distance $2B$ apart, with a uniform velocity profile $v_z = v_0$. For $z > 0$, heat is added to the fluid with a constant flux q_0 at both walls. The liquid enters the heated region of the slit at $z = 0$ with a uniform temperature T_0 .

Answer the questions in terms of the following parameters (all assumed constant):

v_0 : velocity of the fluid in z -direction

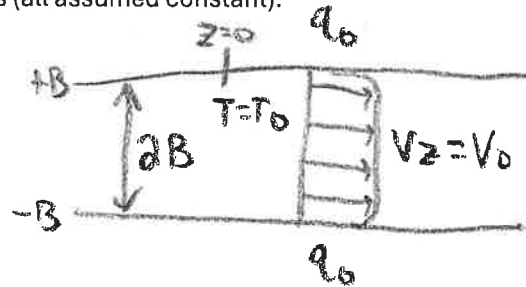
B : half the distance between the two parallel walls

k : thermal conductivity of the fluid

ρ : density of the fluid

c_p : isobaric heat capacity per unit mass of the fluid

q_0 : heat flux at the walls (assumed positive)



- 15/15 2.1. Write down a steady state macroscopic energy balance on the entire fluid between 0 and z to find the bulk average temperature as a function of z . Assume constant heat capacity and neglect the pressure dependence of enthalpy.

$$\int_V dV \frac{\partial}{\partial t} \rho \left(\frac{1}{2} v^2 + \phi + u \right) = - \int_S dS \rho \left(\frac{1}{2} v^2 + \phi + u \right) \langle v|n \rangle - \int_S dS \langle q|n \rangle - \int_S dS \langle v|\tau|n \rangle - \int_S dS p \langle v|n \rangle$$

$$T_b(z) \equiv \frac{\int_A dS v_z T}{\int_A dS v_z}$$

- 15/15 2.2. Write down a steady state shell energy balance to obtain a partial differential equation (PDE) for the temperature of the liquid as a function of x and z .

For large z , the temperature profile assumes the following asymptotic solution:

$$T(x, z) = T_b(z) + \psi(x)$$

$$\frac{\int_S \psi}{V_2} = 0$$

- 15/15 2.3. Derive an ordinary differential equation (ODE) for $\psi(x)$ by replacing the asymptotic solution in the PDE derived in the previous part. Solve the ODE and evaluate the integration constants.

- 10/15 2.4. Using the asymptotic temperature profile derived above find the Nusselt number.

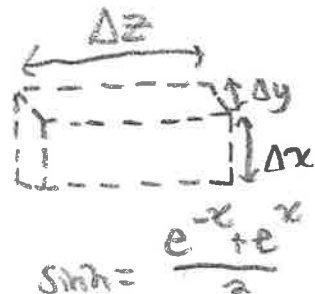
$$Nu = \frac{4hB}{k}$$

$$h[T(B, z) - T_b(z)] = q_0$$

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1.1



$$\sinh = \frac{e^{-x} + e^x}{2} \quad \cosh = \frac{e^{-x} - e^x}{2} - K''' C_A \Delta x \Delta y$$

Mole Balance: $0 = -\int_S \langle N_A n \rangle ds + \int_V R_A dV$

$$\Rightarrow 0 = [\Delta z \Delta y N_{Ax}]_x - [\Delta z \Delta y N_{Ax}]_{x+\Delta x} + [\Delta x \Delta y N_{Az}]_z - [\Delta x \Delta y N_{Az}]_{z+\Delta z}$$

$$\Rightarrow -\frac{\partial N_{Ax}}{\partial x} - K_1''' C_A = 0 \rightarrow N_{Ax} = x_A \sum_{B=1}^N N_B - C_{DAB} \left(\frac{\partial x_A}{\partial x} + \frac{\partial x_A}{\partial z} \right)$$

$$\Rightarrow N_{Ax} = -D_{AB} \left(\frac{\partial C_A}{\partial x} + \frac{\partial C_A}{\partial z} \right) \rightarrow D_{AB} \frac{\partial^2 C_A}{\partial x^2} + D_{AB} \frac{\partial}{\partial x} \left(\frac{\partial C_A}{\partial z} \right) - K_1''' C_A = 0$$

1.2

$$\frac{\partial C_A}{\partial z} = 0 \rightarrow D_{AB} \frac{\partial^2 C_A}{\partial x^2} - K_1''' C_A = 0 \rightarrow \frac{\partial^2 C_A}{\partial x^2} = \frac{K_1'''}{D_{AB}} C_A \rightarrow \frac{\partial^2 C_A}{\partial x^2} = \phi^2 C_A$$

$$\Rightarrow C_A = C_1 \sinh(\phi x) + C_2 \cosh(\phi x) \quad \begin{cases} C_A(x=B) = C_{A0} \\ C_A(x=-B) = C_{A0} \end{cases}$$

$$\Rightarrow C_{A0} = C_1 \sinh(B\phi) + C_2 \cosh(B\phi), \quad C_{A0} = C_1 \sinh(-B\phi) + C_2 \cosh(-B\phi)$$

$$\Rightarrow C_1 = \frac{C_{A0}}{\sinh(B\phi)} - C_2 \coth(B\phi) \rightarrow C_{A0} = C_{A0} \frac{\sinh(-B\phi)}{\sinh(B\phi)} - C_2 \coth(B\phi) \sinh(B\phi) + C_2 \cosh(-B\phi)$$

$$\Rightarrow C_2 \left[\cosh(-B\phi) - \coth(B\phi) \sinh(-B\phi) \right] = C_{A0} \left(1 - \frac{\sinh(-B\phi)}{\sinh(B\phi)} \right)$$

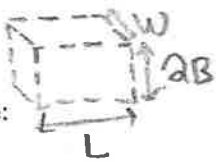
$$\Rightarrow C_2 = \frac{C_{A0} \left(1 - \frac{\sinh(-B\phi)}{\sinh(B\phi)} \right)}{\left[\cosh(-B\phi) - \coth(B\phi) \sinh(-B\phi) \right]}$$

$$\Rightarrow C_1 = \frac{C_{A0}}{\sinh(B\phi)} - \left[\frac{C_{A0} \left(1 - \frac{\sinh(-B\phi)}{\sinh(B\phi)} \right)}{\left[\cosh(-B\phi) - \coth(B\phi) \sinh(-B\phi) \right]} \right] \coth(B\phi)$$

Plug back into C_A to get $C_A = C_A(x)$ at large z

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Left & Right

Top/Bottom

2.1

Macroscopic
Energy Balance

$$0 = \left[\rho \left(\frac{1}{2} V_0^3 + u V_0 \right) \right]_0 - \left[\rho \left(\frac{1}{2} V_0^3 + u V_0 \right) \right]_z + 2 q_0 \cdot W L + 0$$

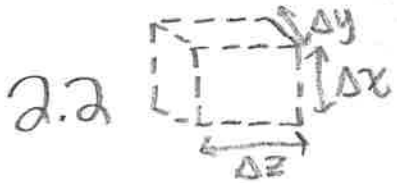
$\cdot 2 B W$ $\cdot 2 B W$ $\cdot W L$

$$\Rightarrow \rho B W V_0^3 - \rho B W V_0^3 + 2 B W V_0 u(T_0) - 2 B W V_0 u(T) + 2 W L q_0 = 0$$

$$\Rightarrow 2 B W V_0 C_p (T_0 - T) + 2 W L q_0 = 0 \rightarrow q_0 = \frac{2 B W V_0 C_p (T - T_0) \rho}{2 W L}$$

$$\Rightarrow q_0 = \frac{V_0 B C_p (T - T_0) \rho}{L} \rightarrow T = T_0 + \frac{q_0 L}{V_0 B C_p \rho} \rightarrow T = T_0 + \frac{q_0}{V_0 B C_p \rho} z$$

$$\Rightarrow T_b = \frac{\int_A V_z \cdot T ds}{\int_A V_z} = \frac{V_0 \int_A T \cdot ds}{V_0 \int_A ds} = \frac{\int_A T ds}{\int_A ds} = T \rightarrow \boxed{T_b = T_0 + \frac{q_0}{V_0 B C_p \rho} z}$$



2.2

$$\Rightarrow 0 = [\rho u V_0 \Delta x \Delta y]_z - [\rho u V_0 \Delta x \Delta y]_{z+\Delta z} + [q \Delta y \Delta z]_x - [q \Delta y \Delta z]_{x+\Delta x}$$

$$\Rightarrow 0 = -\rho V_0 \frac{\partial u}{\partial z} - \frac{\partial q}{\partial x} \rightarrow \boxed{0 = -\rho V_0 C_p \frac{\partial T}{\partial z} + K \frac{\partial^2 T}{\partial x^2}} \quad q_0 = -K \frac{dT}{dx}$$

2.3

$$0 = -\rho V_0 C_p \frac{dT_b}{dz} + K \frac{\partial^2 \psi}{\partial x^2} \rightarrow \frac{\partial^2 \psi}{\partial x^2} = \frac{\rho V_0 C_p}{K} \left(\frac{q_0}{V_0 B C_p \rho} \right)$$

$$\Rightarrow \frac{\partial^2 \psi}{\partial x^2} = \frac{q_0}{K B} \rightarrow \frac{d\psi}{dx} = \frac{q_0}{K B} x + C_1 \rightarrow \psi = \frac{q_0}{2 K B} x^2 + C_1 x + C_2$$

$$\begin{cases} \frac{\partial T}{\partial x}(x=B) = +q_0/K \\ \int_S \psi ds = 0 \end{cases} \rightarrow \frac{+q_0}{K} = \frac{q_0}{K} + C_1 \rightarrow C_1 = 0$$

$$\Rightarrow \int_S \psi ds = W \int_{-B}^B \frac{q_0}{2 K B} x^2 dx + C_2 = 0$$

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$$\Rightarrow W \int_{-B}^B \frac{q_0}{2KB} x^2 - \frac{q_0 B}{K} x + C_2 = 0$$

$$\Rightarrow W \left[\frac{q_0}{6KB} x^3 - \frac{q_0 B}{K} x + C_2 x \right]_{-B}^B = 0$$

$$\Rightarrow W \left[\frac{q_0 B^2}{6K} + \frac{q_0 B^2}{6K} - \frac{q_0 B^2}{K} + \frac{q_0 B^2}{K} + 2C_2 B \right] = 0$$

$$\Rightarrow \frac{W q_0 B^2}{3K} + 2C_2 B = 0 \rightarrow C_2 = -\frac{W q_0 B}{6K} \rightarrow C_2 = -\frac{q_0 B}{6K}$$

$$\Rightarrow \psi(x) = \frac{q_0}{2KB} x^2 - \frac{q_0 B}{6K}$$

$$\Rightarrow T(x, z) = T_0 + \frac{q_0}{V_0 B C_p P} z + \frac{q_0}{2KB} x^2 - \frac{q_0 B}{6K}$$

$$2.4 \quad h(T(x, z) - T_b) = q_0, \quad Nu = \frac{4hB}{K}$$

$$\Rightarrow h \left(\frac{q_0}{2KB} x^2 - \frac{q_0 B}{6K} \right) = q_0 \quad @ x=B \rightarrow h \left(\frac{q_0 B}{2K} - \frac{q_0 B}{6K} \right) = q_0$$

$$\Rightarrow h \left(\frac{3B}{6K} - \frac{B}{6K} \right) = 1 \rightarrow h \left(\frac{2B}{6K} \right) = 1 \rightarrow h = \frac{3K}{B}$$

$$\Rightarrow Nu = \frac{4hB}{K} = \frac{4(3K/B)B}{K} = 12$$

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